

Classical Machine Learning

A Napkin-style companion for campus placements:
derivations, hand numerics, traps, and code that checks itself

Philospher

Built October 6, 2026

Preface

[STUB: preface]

Contents

I	Foundations	9
1	The Setup	11
1.1	The zoo of learning problems	11
1.2	Loss, risk and empirical risk	11
1.3	Train, validation, test	11
1.4	Over- and underfitting, generalisation gap	11
1.5	The i.i.d. assumption and when it breaks	11
2	Maths at Your Fingertips	13
2.1	Linear algebra	13
2.2	Calculus and matrix calculus	13
2.3	Probability	13
2.4	MLE and MAP	13
2.5	Convexity	13
2.6	Lagrange multipliers and KKT	13
II	Supervised Learning	15
3	Linear Regression	19
3.1	Least squares and the normal equations	19
3.2	The projection and MLE views	19
3.3	Five points by hand	19
3.4	R-squared and adjusted R-squared	19
3.5	Assumptions, diagnostics, multicollinearity	19
3.6	Polynomial features	19
4	Bias, Variance and Regularisation	21
4.1	The decomposition	21
4.2	Ridge	21
4.3	Lasso and elastic net	21
4.4	The augmented error	21
4.5	Early stopping and double descent	21
5	Logistic Regression and GLMs	23
5.1	From odds to the sigmoid	23
5.2	Log-loss from MLE	23
5.3	Newton and IRLS	23
5.4	Softmax regression and multiclass	23
5.5	Imbalance and thresholds	23
5.6	The perceptron and XOR	23
6	Generative Classifiers	25
6.1	Naive Bayes	25
6.2	Laplace smoothing	25
6.3	LDA and QDA	25

7	k-Nearest Neighbours and Distances	27
7.1	The algorithm	27
7.2	Distances	27
7.3	Choosing k	27
7.4	The curse of dimensionality	27
7.5	KD-trees and approximate search	27
8	Support Vector Machines	29
8.1	Margins	29
8.2	Hard and soft margin	29
8.3	Duality	29
8.4	Kernels	29
8.5	SVR and SMO	29
8.6	Convex surrogate losses	29
9	Decision Trees	31
9.1	Impurity	31
9.2	A split by hand	31
9.3	Growing a tree	31
9.4	Regression trees	31
9.5	Pruning	31
9.6	What trees are good for	31
10	Ensembles	33
10.1	The bootstrap and 63.2 percent	33
10.2	Bagging and random forests	33
10.3	Feature importance and its traps	33
10.4	AdaBoost	33
10.5	Gradient boosting	33
10.6	XGBoost	33
10.7	LightGBM and CatBoost	33
10.8	Stacking and the bias-variance story	33
III	Learning Without (Enough) Labels	35
11	Unsupervised Learning	37
11.1	k-means	37
11.2	Hierarchical clustering	37
11.3	DBSCAN	37
11.4	Gaussian mixtures and EM	37
11.5	PCA	37
11.6	ICA, t-SNE and UMAP	37
11.7	Anomaly detection	37
11.8	Association rules	37
12	Semi- and Self-Supervised Learning	39
12.1	Assumptions	39
12.2	Self-training	39
12.3	Co-training and label propagation	39
12.4	Consistency regularisation	39

12.5	When unlabelled data hurts	39
12.6	Contrastive learning in one page	39
IV	Doing It Properly	41
13	Evaluation and Model Selection	43
13.1	The confusion matrix	43
13.2	ROC and PR curves	43
13.3	Regression and probabilistic metrics	43
13.4	Cross-validation	43
13.5	Leakage	43
13.6	AIC, BIC and hyper-parameter search	43
13.7	Calibration	43
14	Statistics Interviewers Ask	45
14.1	Distributions	45
14.2	Estimators and confidence intervals	45
14.3	Hypothesis tests	45
14.4	Chi-square and ANOVA	45
14.5	Multiple testing	45
14.6	Correlation, causation and Simpson	45
15	Data Preparation and Feature Engineering	47
15.1	Scaling	47
15.2	Encoding categoricals	47
15.3	Missing values and outliers	47
15.4	Feature selection	47
15.5	Imbalance	47
15.6	Text features and TF-IDF	47
15.7	Time series	47
15.8	Recommenders	47
16	Optimisation for ML	49
16.1	Gradient descent on a quadratic	49
16.2	Momentum	49
16.3	Newton and quasi-Newton	49
16.4	Coordinate descent	49
16.5	Duality	49
V	Interview Mode	51
17	Implement It	53
17.1	Linear models	53
17.2	Neighbours and Bayes	53
17.3	Trees and forests	53
17.4	Boosting	53
17.5	Clustering and PCA	53
17.6	SVM	53

18	Rapid-fire	55
18.1	Rapid-fire answers	55
19	Problems	57
19.1	Problems	57
VI	Backmatter	59
A	Hints to problems	61
B	Solutions to problems	63

I

Foundations

Part I: Contents

1	The Setup	11
1.1	The zoo of learning problems	11
1.2	Loss, risk and empirical risk	11
1.3	Train, validation, test	11
1.4	Over- and underfitting, generalisation gap	11
1.5	The i.i.d. assumption and when it breaks	11
2	Maths at Your Fingertips	13
2.1	Linear algebra	13
2.2	Calculus and matrix calculus	13
2.3	Probability	13
2.4	MLE and MAP	13
2.5	Convexity	13
2.6	Lagrange multipliers and KKT	13

1 The Setup

§1.1 The zoo of learning problems

[STUB: The zoo of learning problems]

§1.2 Loss, risk and empirical risk

[STUB: Loss, risk and empirical risk]

§1.3 Train, validation, test

[STUB: Train, validation, test]

§1.4 Over- and underfitting, generalisation gap

[STUB: Over- and underfitting, generalisation gap]

§1.5 The i.i.d. assumption and when it breaks

[STUB: The i.i.d. assumption and when it breaks]

2 Maths at Your Fingertips

§2.1 Linear algebra

[STUB: Linear algebra]

§2.2 Calculus and matrix calculus

[STUB: Calculus and matrix calculus]

§2.3 Probability

[STUB: Probability]

§2.4 MLE and MAP

[STUB: MLE and MAP]

§2.5 Convexity

[STUB: Convexity]

§2.6 Lagrange multipliers and KKT

[STUB: Lagrange multipliers and KKT]

II

Supervised Learning

Part II: Contents

3	Linear Regression	19
3.1	Least squares and the normal equations	19
3.2	The projection and MLE views	19
3.3	Five points by hand	19
3.4	R-squared and adjusted R-squared	19
3.5	Assumptions, diagnostics, multicollinearity	19
3.6	Polynomial features	19
4	Bias, Variance and Regularisation	21
4.1	The decomposition	21
4.2	Ridge	21
4.3	Lasso and elastic net	21
4.4	The augmented error	21
4.5	Early stopping and double descent	21
5	Logistic Regression and GLMs	23
5.1	From odds to the sigmoid	23
5.2	Log-loss from MLE	23
5.3	Newton and IRLS	23
5.4	Softmax regression and multiclass	23
5.5	Imbalance and thresholds	23
5.6	The perceptron and XOR	23
6	Generative Classifiers	25
6.1	Naive Bayes	25
6.2	Laplace smoothing	25
6.3	LDA and QDA	25
7	k-Nearest Neighbours and Distances	27
7.1	The algorithm	27
7.2	Distances	27
7.3	Choosing k	27
7.4	The curse of dimensionality	27
7.5	KD-trees and approximate search	27
8	Support Vector Machines	29
8.1	Margins	29
8.2	Hard and soft margin	29
8.3	Duality	29
8.4	Kernels	29
8.5	SVR and SMO	29
8.6	Convex surrogate losses	29
9	Decision Trees	31
9.1	Impurity	31
9.2	A split by hand	31
9.3	Growing a tree	31
9.4	Regression trees	31
9.5	Pruning	31
9.6	What trees are good for	31
10	Ensembles	33
10.1	The bootstrap and 63.2 percent	33
10.2	Bagging and random forests	33
10.3	Feature importance and its traps	33

10.4	AdaBoost	33
10.5	Gradient boosting	33
10.6	XGBoost	33
10.7	LightGBM and CatBoost	33
10.8	Stacking and the bias-variance story	33

3 Linear Regression

§3.1 Least squares and the normal equations

[STUB: Least squares and the normal equations]

§3.2 The projection and MLE views

[STUB: The projection and MLE views]

§3.3 Five points by hand

[STUB: Five points by hand]

§3.4 R-squared and adjusted R-squared

[STUB: R-squared and adjusted R-squared]

§3.5 Assumptions, diagnostics, multicollinearity

[STUB: Assumptions, diagnostics, multicollinearity]

§3.6 Polynomial features

[STUB: Polynomial features]

4 Bias, Variance and Regularisation

§4.1 The decomposition

[STUB: The decomposition]

§4.2 Ridge

[STUB: Ridge]

§4.3 Lasso and elastic net

[STUB: Lasso and elastic net]

§4.4 The augmented error

[STUB: The augmented error]

§4.5 Early stopping and double descent

[STUB: Early stopping and double descent]

5 Logistic Regression and GLMs

§5.1 From odds to the sigmoid

[STUB: From odds to the sigmoid]

§5.2 Log-loss from MLE

[STUB: Log-loss from MLE]

§5.3 Newton and IRLS

[STUB: Newton and IRLS]

§5.4 Softmax regression and multiclass

[STUB: Softmax regression and multiclass]

§5.5 Imbalance and thresholds

[STUB: Imbalance and thresholds]

§5.6 The perceptron and XOR

[STUB: The perceptron and XOR]

6 Generative Classifiers

§6.1 Naive Bayes

[STUB: Naive Bayes]

§6.2 Laplace smoothing

[STUB: Laplace smoothing]

§6.3 LDA and QDA

[STUB: LDA and QDA]

7 k-Nearest Neighbours and Distances

§7.1 The algorithm

[STUB: The algorithm]

§7.2 Distances

[STUB: Distances]

§7.3 Choosing k

[STUB: Choosing k]

§7.4 The curse of dimensionality

[STUB: The curse of dimensionality]

§7.5 KD-trees and approximate search

[STUB: KD-trees and approximate search]

8 Support Vector Machines

§8.1 Margins

[STUB: Margins]

§8.2 Hard and soft margin

[STUB: Hard and soft margin]

§8.3 Duality

[STUB: Duality]

§8.4 Kernels

[STUB: Kernels]

§8.5 SVR and SMO

[STUB: SVR and SMO]

§8.6 Convex surrogate losses

[STUB: Convex surrogate losses]

9 Decision Trees

§9.1 Impurity

[STUB: Impurity]

§9.2 A split by hand

[STUB: A split by hand]

§9.3 Growing a tree

[STUB: Growing a tree]

§9.4 Regression trees

[STUB: Regression trees]

§9.5 Pruning

[STUB: Pruning]

§9.6 What trees are good for

[STUB: What trees are good for]

10 Ensembles

§10.1 The bootstrap and 63.2 percent

[STUB: The bootstrap and 63.2 percent]

§10.2 Bagging and random forests

[STUB: Bagging and random forests]

§10.3 Feature importance and its traps

[STUB: Feature importance and its traps]

§10.4 AdaBoost

[STUB: AdaBoost]

§10.5 Gradient boosting

[STUB: Gradient boosting]

§10.6 XGBoost

[STUB: XGBoost]

§10.7 LightGBM and CatBoost

[STUB: LightGBM and CatBoost]

§10.8 Stacking and the bias-variance story

[STUB: Stacking and the bias-variance story]

III

Learning Without (Enough) Labels

Part III: Contents

11	Unsupervised Learning	37
11.1	k-means	37
11.2	Hierarchical clustering	37
11.3	DBSCAN	37
11.4	Gaussian mixtures and EM	37
11.5	PCA	37
11.6	ICA, t-SNE and UMAP	37
11.7	Anomaly detection	37
11.8	Association rules	37
12	Semi- and Self-Supervised Learning	39
12.1	Assumptions	39
12.2	Self-training	39
12.3	Co-training and label propagation	39
12.4	Consistency regularisation	39
12.5	When unlabelled data hurts	39
12.6	Contrastive learning in one page	39

11 Unsupervised Learning

§11.1 k-means

[STUB: k-means]

§11.2 Hierarchical clustering

[STUB: Hierarchical clustering]

§11.3 DBSCAN

[STUB: DBSCAN]

§11.4 Gaussian mixtures and EM

[STUB: Gaussian mixtures and EM]

§11.5 PCA

[STUB: PCA]

§11.6 ICA, t-SNE and UMAP

[STUB: ICA, t-SNE and UMAP]

§11.7 Anomaly detection

[STUB: Anomaly detection]

§11.8 Association rules

[STUB: Association rules]

12 Semi- and Self-Supervised Learning

§12.1 Assumptions

[STUB: Assumptions]

§12.2 Self-training

[STUB: Self-training]

§12.3 Co-training and label propagation

[STUB: Co-training and label propagation]

§12.4 Consistency regularisation

[STUB: Consistency regularisation]

§12.5 When unlabelled data hurts

[STUB: When unlabelled data hurts]

§12.6 Contrastive learning in one page

[STUB: Contrastive learning in one page]

IV

Doing It Properly

Part IV: Contents

13	Evaluation and Model Selection	43
13.1	The confusion matrix	43
13.2	ROC and PR curves	43
13.3	Regression and probabilistic metrics	43
13.4	Cross-validation	43
13.5	Leakage	43
13.6	AIC, BIC and hyper-parameter search	43
13.7	Calibration	43
14	Statistics Interviewers Ask	45
14.1	Distributions	45
14.2	Estimators and confidence intervals	45
14.3	Hypothesis tests	45
14.4	Chi-square and ANOVA	45
14.5	Multiple testing	45
14.6	Correlation, causation and Simpson	45
15	Data Preparation and Feature Engineering	47
15.1	Scaling	47
15.2	Encoding categoricals	47
15.3	Missing values and outliers	47
15.4	Feature selection	47
15.5	Imbalance	47
15.6	Text features and TF-IDF	47
15.7	Time series	47
15.8	Recommenders	47
16	Optimisation for ML	49
16.1	Gradient descent on a quadratic	49
16.2	Momentum	49
16.3	Newton and quasi-Newton	49
16.4	Coordinate descent	49
16.5	Duality	49

13 Evaluation and Model Selection

§13.1 The confusion matrix

[STUB: The confusion matrix]

§13.2 ROC and PR curves

[STUB: ROC and PR curves]

§13.3 Regression and probabilistic metrics

[STUB: Regression and probabilistic metrics]

§13.4 Cross-validation

[STUB: Cross-validation]

§13.5 Leakage

[STUB: Leakage]

§13.6 AIC, BIC and hyper-parameter search

[STUB: AIC, BIC and hyper-parameter search]

§13.7 Calibration

[STUB: Calibration]

14 Statistics Interviewers Ask

§14.1 Distributions

[STUB: Distributions]

§14.2 Estimators and confidence intervals

[STUB: Estimators and confidence intervals]

§14.3 Hypothesis tests

[STUB: Hypothesis tests]

§14.4 Chi-square and ANOVA

[STUB: Chi-square and ANOVA]

§14.5 Multiple testing

[STUB: Multiple testing]

§14.6 Correlation, causation and Simpson

[STUB: Correlation, causation and Simpson]

15 Data Preparation and Feature Engineering

§15.1 Scaling

[STUB: Scaling]

§15.2 Encoding categoricals

[STUB: Encoding categoricals]

§15.3 Missing values and outliers

[STUB: Missing values and outliers]

§15.4 Feature selection

[STUB: Feature selection]

§15.5 Imbalance

[STUB: Imbalance]

§15.6 Text features and TF-IDF

[STUB: Text features and TF-IDF]

§15.7 Time series

[STUB: Time series]

§15.8 Recommenders

[STUB: Recommenders]

16 Optimisation for ML

§16.1 Gradient descent on a quadratic

[STUB: Gradient descent on a quadratic]

§16.2 Momentum

[STUB: Momentum]

§16.3 Newton and quasi-Newton

[STUB: Newton and quasi-Newton]

§16.4 Coordinate descent

[STUB: Coordinate descent]

§16.5 Duality

[STUB: Duality]



Interview Mode

Part V: Contents

17	Implement It	53
17.1	Linear models	53
17.2	Neighbours and Bayes	53
17.3	Trees and forests	53
17.4	Boosting	53
17.5	Clustering and PCA	53
17.6	SVM	53
18	Rapid-fire	55
18.1	Rapid-fire answers	55
19	Problems	57
19.1	Problems	57

17 Implement It

§17.1 Linear models

[STUB: Linear models]

§17.2 Neighbours and Bayes

[STUB: Neighbours and Bayes]

§17.3 Trees and forests

[STUB: Trees and forests]

§17.4 Boosting

[STUB: Boosting]

§17.5 Clustering and PCA

[STUB: Clustering and PCA]

§17.6 SVM

[STUB: SVM]

18 Rapid-fire

§18.1 Rapid-fire answers

[STUB: Rapid-fire answers]

19 Problems

§19.1 Problems

[STUB: Problems]

VI

Backmatter

A Hints to problems

B Solutions to problems